Subject: Mathematics **WEEKLY CHAPTER TEST – 05 (PROBABILITY)** **Time:** 45 Minutes
Class: Class X (CBSE) **Chapter 14: Probability** **MM: 20**

Student Name: _____ **Roll No.:** _____ **Date:** _____

GENERAL INSTRUCTIONS:

1. All questions are compulsory. Section A has 4 MCQs and 1 Assertion-Reason question (1 mark each).
2. Section B has 1 VSAQ question (2 marks). Section C has 3 SA questions (3 marks each).
3. Section D has 1 Applied Case Study (4 marks). Show all working clearly.

SECTION A: OBJECTIVE & MCQs (1 Mark Each) [5 Marks]

1. In a single throw of two dice, the probability of getting 6 as a product of the two numbers obtained is: **[1]**

Options: (a) $\frac{1}{9}$ (b) $\frac{2}{9}$ (c) $\frac{1}{6}$ (d) $\frac{5}{36}$

2. The king, queen, and jack of clubs are removed from a deck of 52 cards. A card is drawn from the remaining cards. The probability of getting a club card is: **[1]**

Options: (a) $\frac{10}{49}$ (b) $\frac{13}{49}$ (c) $\frac{1}{4}$ (d) $\frac{3}{49}$

3. A box contains 5 apples, 6 oranges, and x bananas. If the probability of selecting an apple from the box is $\frac{1}{3}$, then the number of bananas in the box is: **[1]**

Options: (a) 3 (b) 4 (c) 5 (d) 6

4. A letter is chosen at random from the letters of the word 'ASSASSINATION'. The probability that the letter chosen is a vowel is: **[1]**

Options: (a) $\frac{6}{13}$ (b) $\frac{7}{13}$ (c) $\frac{5}{13}$ (d) $\frac{4}{13}$

5. **Assertion (A):** When two coins are tossed simultaneously, the probability of getting at least one head is $\frac{3}{4}$.
Reason (R): The sample space of tossing two coins is $\{HH, HT, TH, TT\}$, containing 4 equally likely outcomes. **[1]**

(a) Both (A) and (R) are true, and (R) is the correct explanation of (A).
(b) Both (A) and (R) are true, but (R) is not the correct explanation of (A).
(c) (A) is true, but (R) is false.
(d) (A) is false, but (R) is true.

SECTION B: VERY SHORT ANSWER QUESTIONS (2 Marks Each) [2 Marks]

6. A die is thrown once. Find the probability of getting: (i) a number divisible by 3 (ii) a composite number. **[2]**

SECTION C: SHORT ANSWER QUESTIONS (3 Marks Each) [9 Marks]

7. A game consists of tossing a one-rupee coin 3 times and noting its outcome each time. Hanif wins if all the tosses give the same result (i.e., three heads or three tails), and loses otherwise. (i) Write all possible outcomes of the game. (ii) Calculate the probability that Hanif will lose the game. (iii) What is the probability that he gets at least one head? **[3]**

8. A bag contains 6 red, 5 black, and 4 white balls. A ball is drawn from the bag at random. Find the probability that the ball drawn is: (i) white (ii) red or black (iii) neither red nor black. **[3]**

9. Two customers, Shyam and Ekta, are visiting a particular shop in the same week (Tuesday to Saturday). Each is equally likely to visit the shop on any day as on another day. What is the probability that both will visit the shop on: (i) the same day? (ii) consecutive days? (iii) different days? **[3]**

SECTION D: APPLIED CASE STUDY (4 Marks) [4 Marks]

Question 10: Case Study Analysis

A school fair organizer sets up a game called "Lucky Trio" involving the simultaneous toss of three fair coins. The entry fee is Rs. 200. The payout rules are structured as follows:

- If a player gets three heads, they win double their entry fee (Rs. 400).
- If a player gets exactly two heads, their entry fee of Rs. 200 is refunded.
- If a player gets one head or zero heads, their entry fee is forfeited (they lose). Shikha enters the game by paying Rs. 200.

Based on the above scenario, answer the following:

(i) Write down all 8 outcomes in the sample space. **[1]**

(ii) What is the probability that Shikha wins double the entry fee? **[1]**

(iii) What is the probability that Shikha gets her entry fee refunded? **[1]**

(iv) What is the probability that Shikha loses her entry fee? **[1]**

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